Decision-Making Under Uncertainty: Exam Solutions

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Learning objective

Structure decision-making problems under uncertainty, calculate and organise payoffs, and apply the Maximax and Maximin criteria to compare alternatives and justify decisions.

Criteria assessment

This task sheet assesses criteria C1 through C4. Each problem assesses the criteria indicated in its title. The number of points assigned to each task is shown at the end of the item. Partial credit may be awarded when the correct procedure is shown but arithmetic errors are made. Answers that do not show the required procedure will not receive credit. A minimum of 4 points is required to pass each criterion.

Evaluated criteria

C1 Characterises decision-making problems under uncertainty by identifying decision alternatives, states of nature, consequences, and the economic or financial context of the decision.

C2 Organises decision problems using payoff tables, placing alternatives, states of nature, and payoffs in a clear structure for comparison.

C3 Calculates profits by combining revenues with fixed and variable costs across different states of nature, determining the corresponding payoffs for each alternative, and organising the results in payoff tables for comparison.

C4 Applies the Maximin and Maximax criteria to make optimistic or conservative decisions without probabilities, selecting the alternative with the highest possible payoff or the best worst-case, interpreting risks or limitations.

1 Meal Delivery Service [C1-1 C2-2]

A meal delivery company is planning its service offering for an upcoming operating period. A standard delivery service is expected to generate profits of twenty thousand US dollars under high demand and twelve thousand US dollars under low demand, while a premium delivery service is expected to generate thirty-four thousand US dollars and seven thousand US dollars, respectively. The company must select its delivery service before the actual level of customer demand is known.

Questions/Tasks.

C1 Characterize the key elements of the decision-making situation. **[1]**
C2 Construct the payoff table. **[2]**

C1 - Characterization [1]

Element	Description	
Alternatives	A_1: Standard delivery service	A_2: Premium delivery service
States of nature	S_1: High demand	S_2: Low demand
Uncertain event	Actual customer demand after the delivery service decision	
Consequences	Operating the selected activity under the realised conditions leaves funds available or a shortfall to cover. Monetary measure: Profit in thousand US dollars for each alternative–state pair	

C2 - Payoff table [2]

Each supplied monetary payoff is placed at the intersection of its alternative and state:

$$P(A_1, S_1) = 20, \quad P(A_1, S_2) = 12,$$
$$P(A_2, S_1) = 34, \quad P(A_2, S_2) = 7.$$

Thus, in thousand US dollars, the completed payoff table is

Alternative	High S_1	Low S_2
Standard delivery service A_1	20	12
Premium delivery service A_2	34	7



2 Small Coffee Roastery Production [C1-1 C2-3]

A small coffee roastery is planning its production capacity for an upcoming sales period. A standard roasting machine can generate 4 thousand BRL per production batch, while an industrial roasting machine can generate 6 thousand BRL per

batch. Based on expected demand, the standard roasting machine would complete 18 batches under high demand or 12 under low demand, whereas the industrial roasting machine would complete 15 or 10 batches, respectively. In addition, the roastery would receive a fixed sales bonus of 7 thousand BRL if demand is high and incur a fixed loss of 4 thousand BRL if demand is low.

Questions/Tasks.

C1 Characterize the key elements of the decision-making situation. **[1]**
C2 Construct the payoff table. **[3]**

C1 - Characterization [1]

Element	Description	
Alternatives	A_1**:** Standard roasting machine	A_2**:** Industrial roasting machine
States of nature	S_1**:** High demand	S_2**:** Low demand
Uncertain event	Actual customer demand after the roasting machine decision	
Consequences	Operating the selected activity under the realised conditions leaves funds available or a shortfall to cover. The monetary measure is profit in thousand BRL for each alternative–state pair.	

C2 - Payoff table [3]

Symbols, values, and meanings.

Symbol	Value (including units)	Meaning in this problem
A_i	A_1: Standard roasting machine; A_2: Industrial roasting machine	Alternative labels, in row order.
S_j	S_1: High demand; S_2: Low demand	State labels, in column order; revealed after the choice.
$q(A_i, S_j)$	Standard–high: 18 batches; Standard–low: 12 batches; Industrial–high: 15 batches; Industrial–low: 10 batches	Expected completed production batches for each roasting machine under each demand condition.
$v^A(A_i)$	Standard roasting machine: 4 thousand BRL/batch; Industrial roasting machine: 6 thousand BRL/batch	Earnings per completed production batch from the chosen roasting machine.
$f^S(S_j)$	High demand: 7 thousand BRL; Low demand: -4 thousand BRL	Fixed sales bonus or loss associated with the demand condition.

Payoff equation.

$$P_{ay}(A_i, S_j) = q(A_i, S_j)v^A(A_i) + f^S(S_j).$$

Additive payoff

Symbolic calculations.

Alternative	S_1	S_2
A_1	$q(A_1, S_1) \times v^A(A_1)$ $+ f^S(S_1)$ $= P_{ay}(A_1, S_1)$	$q(A_1, S_2) \times v^A(A_1)$ $+ f^S(S_2)$ $= P_{ay}(A_1, S_2)$
A_2	$q(A_2, S_1) \times v^A(A_2)$ $+ f^S(S_1)$ $= P_{ay}(A_2, S_1)$	$q(A_2, S_2) \times v^A(A_2)$ $+ f^S(S_2)$ $= P_{ay}(A_2, S_2)$

Numerical calculations.

Alternative	S_1	S_2
A_1	18×4 $+ 7$ $= 79$	12×4 $- 4$ $= 44$
A_2	15×6 $+ 7$ $= 97$	10×6 $- 4$ $= 56$

Negative rates reduce the per-unit contribution; negative fixed amounts represent losses or commitments deducted once. A negative final payoff is a shortfall.

Final payoff table (profit in thousand BRL).

Alternative	High demand (S_1)	Low demand (S_2)
Standard roasting machine (A_1)	79	44
Industrial roasting machine (A_2)	97	56

3 Online Fitness Membership [C1-2 C2-5]

An online fitness company must choose between a standard membership and a flexible membership before customer demand is known. Demand may be high or low. The standard membership generates 5 thousand BRL per active member and also provides a fixed operating contribution of 5 thousand BRL. The flexible membership generates 7 thousand BRL per active member and provides a fixed operating contribution of 8 thousand BRL. High demand adds another 3 thousand BRL per active member to either membership, while low demand adds 1 thousand BRL per active member. The standard membership is expected to have 15 active members under high demand and 9 under low demand; the flexible membership is expected to have 12 active members under high demand and 7 under low demand.

Questions/Tasks.

C1 Characterize the key elements of the decision-making situation. **[2]**

C2 Construct the payoff table. **[5]**

C1 - Characterization [2]

Element	Description	
Alternatives	A_1: Standard membership	A_2: Flexible membership
States of nature	S_1: High customer demand	S_2: Low customer demand
Uncertain event	Actual customer demand after the membership is selected	
Consequences	Operating the selected activity under the realised conditions leaves funds available or a shortfall to cover. The monetary measure is profit in thousand BRL for each alternative–state pair.	

C2 - Payoff table [5]

Symbols, values, and meanings.

Symbol	Value (including units)	Meaning in this problem
A_i	A_1: Standard membership; A_2: Flexible membership	Alternative labels, in row order.
S_j	S_1: High customer demand; S_2: Low customer demand	State labels, in column order; revealed after the choice.
$q(A_i, S_j)$	Standard–high: 15 members; Standard–low: 9 members; Flexible–high: 12 members; Flexible–low: 7 members	Expected active members for each membership under each level of customer demand.
$v^A(A_i)$	Standard membership: 5 thousand BRL/member; Flexible membership: 7 thousand BRL/member	Contribution per active member from the chosen membership.
$v^S(S_j)$	High demand: 3 thousand BRL/member; Low demand: 1 thousand BRL/member	Additional contribution per active member from customer demand.
$f^A(A_i)$	Standard membership: 5 thousand BRL; Flexible membership: 8 thousand BRL	Fixed operating contribution from the chosen membership.

Payoff equation.

$$P_{ay}(A_i, S_j) = q(A_i, S_j)\left[v^A(A_i) + v^S(S_j)\right] + f^A(A_i).$$

Additive payoff

Symbolic calculations.

Alternative	S_1	S_2
A_1	$q(A_1,S_1)\times$ $[v^A(A_1)+v^S(S_1)]$ $+f^A(A_1)$ $=P_{ay}(A_1,S_1)$	$q(A_1,S_2)\times$ $[v^A(A_1)+v^S(S_2)]$ $+f^A(A_1)$ $=P_{ay}(A_1,S_2)$
A_2	$q(A_2,S_1)\times$ $[v^A(A_2)+v^S(S_1)]$ $+f^A(A_2)$ $=P_{ay}(A_2,S_1)$	$q(A_2,S_2)\times$ $[v^A(A_2)+v^S(S_2)]$ $+f^A(A_2)$ $=P_{ay}(A_2,S_2)$

Numerical calculations.

Alternative	S_1	S_2
A_1	$15\times$ $[5+3]$ $+5$ $=125$	$9\times$ $[5+1]$ $+5$ $=59$
A_2	$12\times$ $[7+3]$ $+8$ $=128$	$7\times$ $[7+1]$ $+8$ $=64$

Final payoff table (profit in thousand BRL).

Alternative	High customer demand (S_1)	Low customer demand (S_2)
Standard membership (A_1)	125	59
Flexible membership (A_2)	128	64

4 Neighborhood Bakery Expansion [C1-1 C3-2]

A neighborhood bakery must decide whether to introduce a breakfast menu or maintain its current product selection before customer traffic is known. Traffic may be strong or weak. All values are in million BRL. Introducing the breakfast menu produces revenue of 68 under strong traffic and 38 under weak traffic, while maintaining the current selection produces revenue of 59 under strong traffic and 49 under weak traffic. The corresponding costs are 23 and 21 for the breakfast menu under strong and weak traffic, respectively, and 24 and 22 for the current selection under strong and weak traffic, respectively. **Questions/Tasks.**

C1 Characterize the key elements of the decision-making situation. **[1]**

C3 Use revenue and costs to construct the payoff table. **[2]**

C1 - Characterization [1]

Element	Description	
Alternatives	A_1: Introduce breakfast menu	A_2: Maintain current selection
States of nature	S_1: Strong traffic	S_2: Weak traffic
Uncertain event	Actual customer traffic after the product selection decision	
Consequences	Operating the selected activity under the realised conditions leaves funds available or a shortfall to cover. The monetary measure is profit in million BRL for each alternative–state pair.	

C3 - Calculating profit and building the payoff table [2]

Symbols, values, and meanings.

Symbol	Value (including units)	Meaning in this problem
A_i	A_1: Introduce breakfast menu; A_2: Maintain current selection	Alternative labels, in row order.
S_j	S_1: Strong traffic; S_2: Weak traffic	State labels, in column order; revealed after the choice.
$R(A_i, S_j)$	Breakfast menu–strong: 68 million BRL; Breakfast menu–weak: 38 million BRL; Current selection–strong: 59 million BRL; Current selection–weak: 49 million BRL	Total revenue generated by each product selection under each traffic condition.
$C(A_i, S_j)$	Breakfast menu–strong: 23 million BRL; Breakfast menu–weak: 21 million BRL; Current selection–strong: 24 million BRL; Current selection–weak: 22 million BRL	Total cost associated with each product selection under each traffic condition.

Profit equation.

$$P(A_i, S_j) = R(A_i, S_j) - C(A_i, S_j).$$

Profit

Symbolic calculations.

Alternative	S_1	S_2
A_1	$R(A_1,S_1) - C(A_1,S_1) = P(A_1,S_1)$	$R(A_1,S_2) - C(A_1,S_2) = P(A_1,S_2)$
A_2	$R(A_2,S_1) - C(A_2,S_1) = P(A_2,S_1)$	$R(A_2,S_2) - C(A_2,S_2) = P(A_2,S_2)$

Numerical calculations.

Alternative	S_1	S_2
A_1	$68-23 = 45$	$38-21 = 17$
A_2	$59-24 = 35$	$49-22 = 27$

Final payoff table (profit in million BRL).

Alternative	Strong traffic (S_1)	Weak traffic (S_2)
Introduce breakfast menu (A_1)	45	17
Maintain current selection (A_2)	35	27

5 Bakery Product Production [C1-1 C3-3]

A bakery is planning its production for an upcoming holiday season. Revenue per production batch is 20 thousand EUR for cakes and 16 thousand EUR for pastries. Under high customer demand, the bakery expects to produce 9 batches of cakes and 6 batches of pastries, while under low customer demand it expects to produce 5 batches of cakes and 4 batches of pastries. The cost per production batch is 8 thousand EUR under high customer demand and 11 thousand EUR under low customer demand.

Questions/Tasks.

C1 Characterize the key elements of the decision-making situation. **[1]**

C3 Use revenue and costs to construct the payoff table. **[3]**

C1 - Characterization [1]

Element	Description	
Alternatives	A_1: Produce cakes	A_2: Produce pastries
States of nature	S_1: High customer demand	S_2: Low customer demand
Uncertain event	Customer demand revealed after the production decision	
Consequences	Operating the selected activity under the realised conditions leaves funds available or a shortfall to cover. The monetary measure is profit in thousand EUR for each product–demand-state pair.	

C3 - Profit and payoff table [3]

Symbols, values, and meanings.

Symbol	Value (including units)	Meaning in this problem
A_i	A_1: Produce cakes; A_2: Produce pastries	Alternative labels, in row order.
S_j	S_1: High customer demand; S_2: Low customer demand	State labels, in column order; revealed after the choice.
$q(A_i, S_j)$	Cakes–high: 9; Cakes–low: 5; Pastries–high: 6; Pastries–low: 4	Production batches for each product under each demand condition.
$r(A_i)$	Cakes: 20 thousand EUR/batch; Pastries: 16 thousand EUR/batch	Revenue per production batch.
$c(S_j)$	High customer demand: 8 thousand EUR/batch; Low customer demand: 11 thousand EUR/batch	Cost per production batch.

Profit equation.

$$R = q(A_i, S_j)r(A_i), \qquad C = q(A_i, S_j)c(S_j), \qquad P(A_i, S_j) = R - C.$$

Revenue

Symbolic calculations.

Alternative	S_1	S_2
A_1	$q(A_1, S_1) \times r(A_1)$ $= R(A_1, S_1)$	$q(A_1, S_2) \times r(A_1)$ $= R(A_1, S_2)$
A_2	$q(A_2, S_1) \times r(A_2)$ $= R(A_2, S_1)$	$q(A_2, S_2) \times r(A_2)$ $= R(A_2, S_2)$

Numerical calculations.

Alternative	S_1	S_2
A_1	9×20 $= 180$	5×20 $= 100$
A_2	6×16 $= 96$	4×16 $= 64$

Cost

Symbolic calculations.

Alternative	S_1	S_2
A_1	$q(A_1, S_1) \times c(S_1)$ $= C(A_1, S_1)$	$q(A_1, S_2) \times c(S_2)$ $= C(A_1, S_2)$
A_2	$q(A_2, S_1) \times c(S_1)$ $= C(A_2, S_1)$	$q(A_2, S_2) \times c(S_2)$ $= C(A_2, S_2)$

Numerical calculations.

Alternative	S_1	S_2
A_1	9×8 $= 72$	5×11 $= 55$
A_2	6×8 $= 48$	4×11 $= 44$

Profit

Symbolic calculations.

Alternative	S_1	S_2
A_1	$R(A_1, S_1) - C(A_1, S_1)$ $= P(A_1, S_1)$	$R(A_1, S_2) - C(A_1, S_2)$ $= P(A_1, S_2)$
A_2	$R(A_2, S_1) - C(A_2, S_1)$ $= P(A_2, S_1)$	$R(A_2, S_2) - C(A_2, S_2)$ $= P(A_2, S_2)$

Numerical calculations.

Alternative	S_1	S_2
A_1	$180 - 72$ $= 108$	$100 - 55$ $= 45$
A_2	$96 - 48$ $= 48$	$64 - 44$ $= 20$

Final payoff table (profit in thousand EUR).

Alternative	High customer demand (S_1)	Low customer demand (S_2)
Produce cakes (A_1)	108	45
Produce pastries (A_2)	48	20

6 Meal Delivery Service Packages [C1-2 C3-5]

A meal delivery service must choose between a standard meal package and a premium meal package before weekly demand is known. Demand may be high or low. Quantities, in hundreds of meal packages, are 10 and 7 for the standard package and 8 and 5 for the premium package under high and low demand, respectively. The package-based revenue rates are 7 and 12 thousand USD per hundred meal packages for the standard and premium packages. High demand adds a state-based revenue rate of 3 thousand USD per hundred meal packages; low demand adds 1. The package-based cost rates are 3 and 5 thousand USD per hundred meal packages, respectively. The fixed operating costs are 14 thousand USD under high demand and 18 thousand USD under low demand.

Questions/Tasks.

C1 Characterize the key elements of the decision-making situation. **[2]**

C3 Use revenue and costs to construct the payoff table. **[5]**

C1 - Characterization [2]

Element	Description	
Alternatives	A_1: Offer a standard meal package	A_2: Offer a premium meal package
States of nature	S_1: High weekly demand	S_2: Low weekly demand
Uncertain event	Weekly market conditions revealed after the meal-package decision	
Consequences	Operating the selected activity under the realised conditions leaves funds available or a shortfall to cover. The monetary measure is profit in thousand USD for each package–market-state pair.	

C3 - Profit and payoff table [5]

Symbols, values, and meanings.

Symbol	Value (including units)	Meaning in this problem
A_i	A_1: Offer a standard meal package; A_2: Offer a premium meal package	Alternative labels, in row order.
S_j	S_1: High weekly demand; S_2: Low weekly demand	State labels, in column order; revealed after the choice.
$q(A_i, S_j)$	Standard–high: 10; Standard–low: 7; Premium–high: 8; Premium–low: 5 hundreds of meal packages	Meal-package lots for each package under each demand level.
$r^A(A_i)$	Standard package: 7 thousand USD/100 meal packages; Premium package: 12 thousand USD/100 meal packages	Package-based revenue rate per hundred meal packages.
$r^S(S_j)$	High demand: 3 thousand USD/100 meal packages; Low demand: 1 thousand USD/100 meal packages	Additional revenue rate per hundred meal packages from weekly demand.
$c^A(A_i)$	Standard package: 3 thousand USD/100 meal packages; Premium package: 5 thousand USD/100 meal packages	Package-based cost rate per hundred meal packages.
$f_c(S_j)$	High demand: 14 thousand USD; Low demand: 18 thousand USD	Fixed operating cost under each demand condition.

Profit equation.

$$R = q(A_i, S_j)\left[r^A(A_i) + r^S(S_j)\right], \qquad C = q(A_i, S_j)c^A(A_i) + f_c(S_j), \qquad P(A_i, S_j) = R - C.$$

Revenue

Symbolic calculations.

Alternative	S_1	S_2
A_1	$q(A_1, S_1)\times$ $[r^A(A_1)+r^S(S_1)]$ $= R(A_1, S_1)$	$q(A_1, S_2)\times$ $[r^A(A_1)+r^S(S_2)]$ $= R(A_1, S_2)$
A_2	$q(A_2, S_1)\times$ $[r^A(A_2)+r^S(S_1)]$ $= R(A_2, S_1)$	$q(A_2, S_2)\times$ $[r^A(A_2)+r^S(S_2)]$ $= R(A_2, S_2)$

Numerical calculations.

Alternative	S_1	S_2
A_1	$10\times$ $[7+3]$ $=100$	$7\times$ $[7+1]$ $=56$
A_2	$8\times$ $[12+3]$ $=120$	$5\times$ $[12+1]$ $=65$

Cost

Symbolic calculations.

Alternative	S_1	S_2
A_1	$q(A_1, S_1)\times c^A(A_1)$ $+ f_c(S_1)$ $= C(A_1, S_1)$	$q(A_1, S_2)\times c^A(A_1)$ $+ f_c(S_2)$ $= C(A_1, S_2)$
A_2	$q(A_2, S_1)\times c^A(A_2)$ $+ f_c(S_1)$ $= C(A_2, S_1)$	$q(A_2, S_2)\times c^A(A_2)$ $+ f_c(S_2)$ $= C(A_2, S_2)$

Numerical calculations.

Alternative	S_1	S_2
A_1	10×3 $+14$ $=44$	7×3 $+18$ $=39$
A_2	8×5 $+14$ $=54$	5×5 $+18$ $=43$

Profit

Symbolic calculations.

Alternative	S_1	S_2
A_1	$R(A_1,S_1)-C(A_1,S_1)$ $=P(A_1,S_1)$	$R(A_1,S_2)-C(A_1,S_2)$ $=P(A_1,S_2)$
A_2	$R(A_2,S_1)-C(A_2,S_1)$ $=P(A_2,S_1)$	$R(A_2,S_2)-C(A_2,S_2)$ $=P(A_2,S_2)$

Numerical calculations.

Alternative	S_1	S_2
A_1	$100-44$ $=56$	$56-39$ $=17$
A_2	$120-54$ $=66$	$65-43$ $=22$

Final payoff table (profit in thousand USD).

Alternative	High weekly demand (S_1)	Low weekly demand (S_2)
Standard meal package (A_1)	56	17
Premium meal package (A_2)	66	22

7 Solar Energy Investment Strategy [C4-2]

A renewable energy company must choose a strategy for investing in a new market. Energy demand could be high, medium, or low. All payoffs are in millions of dollars.

	Energy Demand		
Alternative	High	Medium	Low
Utility-scale solar farm	82	38	6
Rooftop solar program	76	51	34
Community solar project	97	44	26
Postpone investment	28	28	28

Questions/Tasks.

C4 What would the optimistic decision be (Maximax)? What would the conservative decision be (Maximin)? **[2]**

C4 - Maximax and Maximin [2]



Complete the table using letters first.

	A_1	A_2	A_3	A_4	Decision
Max	M_1	M_2	M_3	M_4	$A^*_{\text{Maximax}} = A_3$
Min	m_1	m_2	m_3	m_4	$A^*_{\text{Maximin}} = A_2$

Replace the letters with their values.

	A_1	A_2	A_3	A_4	Decision
Max	82	76	97	28	$A^*_{\text{Maximax}} = A_3$
Min	6	34	26	28	$A^*_{\text{Maximin}} = A_2$

Maximax

$$\begin{aligned} M_1 &= \max\{82, 38, 6\} = 82 \\ M_2 &= \max\{76, 51, 34\} = 76 \\ M_3 &= \max\{97, 44, 26\} = 97 \\ M_4 &= \max\{28, 28, 28\} = 28 \end{aligned}$$

Maximin

$$\begin{aligned} m_1 &= \min\{82, 38, 6\} = 6 \\ m_2 &= \min\{76, 51, 34\} = 34 \\ m_3 &= \min\{97, 44, 26\} = 26 \\ m_4 &= \min\{28, 28, 28\} = 28 \end{aligned}$$

Therefore, the optimistic decision is

$$A^*_{\text{Maximax}} = A_3 quad(\text{Community solar project})$$

and the conservative decision is

$$A^*_{\text{Maximin}} = A_2 \quad (\text{Rooftop solar program}).$$

8 Restaurant Expansion Strategy [C4-2]

A restaurant company must choose a strategy for expanding a new location. Customer demand could be high, medium, or low. All payoffs are in thousands of dollars.

Alternative	Customer demand		
	High	Medium	Low
Large restaurant	78	36	12
Medium restaurant	68	49	24
Small restaurant	91	47	29
No expansion	61	42	37

Questions/Tasks.

C4 What would the optimistic decision be (Maximax)? What would the conservative decision be (Maximin)? **[2]**

C4 - Maximax and Maximin [2]



Complete the table using letters first.

	A_1	A_2	A_3	A_4	Decision
Max	M_1	M_2	M_3	M_4	$A^*_{\text{Maximax}} = A_3$
Min	m_1	m_2	m_3	m_4	$A^*_{\text{Maximin}} = A_4$

Maximax

$$\begin{aligned} M_1 &= \max\{78, 36, 12\} = 78 \\ M_2 &= \max\{68, 49, 24\} = 68 \\ M_3 &= \max\{91, 47, 29\} = 91 \\ M_4 &= \max\{61, 42, 37\} = 61 \end{aligned}$$

Maximin

$$\begin{aligned} m_1 &= \min\{78, 36, 12\} = 12 \\ m_2 &= \min\{68, 49, 24\} = 24 \\ m_3 &= \min\{91, 47, 29\} = 29 \\ m_4 &= \min\{61, 42, 37\} = 37 \end{aligned}$$

Replace the letters with their values.

	A_1	A_2	A_3	A_4	Decision
Max	78	68	91	61	$A^*_{\text{Maximax}} = A_3$
Min	12	24	29	37	$A^*_{\text{Maximin}} = A_4$

Therefore, the optimistic decision is

$$A^*_{\text{Maximax}} = A_3 \quad \text{(Small restaurant)}$$

and the conservative decision is

$$A^*_{\text{Maximin}} = A_4 \quad \text{(No expansion)}.$$

9 Renewable Energy Investment Strategy [C4-2]

A renewable energy company must choose a strategy for introducing a new solar power system. Energy market conditions could be highly favourable, favourable, stable, or unfavourable. All payoffs are in thousands of dollars.

	Energy market conditions			
Alternative	Highly favourable	Favourable	Stable	Unfavourable
Full-scale solar	132	61	10	-38
Expanded solar	91	67	32	6
Basic solar	65	54	40	22
Solar leasing	51	46	37	30

Questions/Tasks.

C4 What would the optimistic decision be (Maximax)? What would the conservative decision be (Maximin)? **[2]**

C4 - Maximax and Maximin **[2]**

Complete the table using letters first.

	A_1	A_2	A_3	A_4	Decision
Max	M_1	M_2	M_3	M_4	$A^*_{\text{Maximax}} = A_1$
Min	m_1	m_2	m_3	m_4	$A^*_{\text{Maximin}} = A_4$

Maximax

$$\begin{aligned} M_1 &= \max\{132, 61, 10, -38\} = 132 \\ M_2 &= \max\{91, 67, 32, 6\} = 91 \\ M_3 &= \max\{65, 54, 40, 22\} = 65 \\ M_4 &= \max\{51, 46, 37, 30\} = 51 \end{aligned}$$

Maximin

$$\begin{aligned} m_1 &= \min\{132, 61, 10, -38\} = -38 \\ m_2 &= \min\{91, 67, 32, 6\} = 6 \\ m_3 &= \min\{65, 54, 40, 22\} = 22 \\ m_4 &= \min\{51, 46, 37, 30\} = 30 \end{aligned}$$

Replace the letters with their values.

	A_1	A_2	A_3	A_4	Decision
Max	132	91	65	51	$A^*_{\text{Maximax}} = A_1$
Min	-38	6	22	30	$A^*_{\text{Maximin}} = A_4$

Therefore, the optimistic decision is

$$A^*_{\text{Maximax}} = A_1 \quad \text{(Full-scale solar)}$$

and the conservative decision is

$$A^*_{\text{Maximin}} = A_4 \quad \text{(Solar leasing)}.$$

10 Summarized Solutions [C1-C4]

10.1 Meal Delivery Service C1–1 C2–2

Element	Description
Alt.	**A1:** Standard delivery service **A2:** Premium delivery service
States	**S1:** High demand **S2:** Low demand
Events	Actual customer demand after the delivery service decision
Cons.	Profit in thousand US dollars for each alternative–state pair

Alt.	$\mathbf{S_1}$	$\mathbf{S_2}$
$\mathbf{A_1}$	20	12
$\mathbf{A_2}$	34	7

10.2 Small Coffee Roastery Production C1–1 C2–3

Element	Description
Alt.	**A1:** Standard roasting machine **A2:** Industrial roasting machine
States	**S1:** High demand **S2:** Low demand
Events	Actual customer demand after the roasting machine decision
Cons.	Profit in thousand BRL for each alternative–state pair

Alt.	$\mathbf{S_1}$	$\mathbf{S_2}$
$\mathbf{A_1}$	$18(4)+7 = 79$	$12(4)-4 = 44$
$\mathbf{A_2}$	$15(6)+7 = 97$	$10(6)-4 = 56$

10.3 Online Fitness Membership C1–2 C2–5

Element	Description
Alt.	**A1:** Standard membership **A2:** Flexible membership
States	**S1:** High customer demand **S2:** Low customer demand
Events	Actual customer demand after the membership is selected
Cons.	Profit in thousand BRL for each alternative–state pair

Alt.	$\mathbf{S_1}$	$\mathbf{S_2}$
$\mathbf{A_1}$	$15(5+3)+5 = 125$	$9(5+1)+5 = 59$
$\mathbf{A_2}$	$12(7+3)+8 = 128$	$7(7+1)+8 = 64$

10.4 Neighborhood Bakery Expansion C1–1 C3–2

Element	Description
Alt.	**A1:** Introduce breakfast menu **A2:** Maintain current selection
States	**S1:** Strong traffic **S2:** Weak traffic
Events	Actual customer traffic after the product selection decision
Cons.	Profit in million BRL for each alternative–state pair

Alt.	$\mathbf{S_1}$	$\mathbf{S_2}$
$\mathbf{A_1}$	$68-23$ $=45$	$38-21$ $=17$
$\mathbf{A_2}$	$59-24$ $=35$	$49-22$ $=27$

10.5 Bakery Product Production C1–1 C3–3

Element	Description
Alt.	**A1:** Produce cakes **A2:** Produce pastries
States	**S1:** High customer demand **S2:** Low customer demand
Events	Customer demand revealed after the production decision
Cons.	Profit in thousand EUR for each product–demand-state pair

Alt.	$\mathbf{S_1}$	$\mathbf{S_2}$
$\mathbf{A_1}$	$9(20)-9(8)$ $=108$	$5(20)-5(11)$ $=45$
$\mathbf{A_2}$	$6(16)-6(8)$ $=48$	$4(16)-4(11)$ $=20$

10.6 Meal Delivery Service Packages C1–2 C3–5

Element	Description
Alt.	**A1:** Offer a standard meal package **A2:** Offer a premium meal package
States	**S1:** High weekly demand **S2:** Low weekly demand
Events	Weekly market conditions revealed after the meal-package decision
Cons.	Profit in thousand USD for each package–market-state pair

Alt.	$\mathbf{S_1}$	$\mathbf{S_2}$
$\mathbf{A_1}$	$10(7+3)-[10(3)+14]$ $=56$	$7(7+1)-[7(3)+18]$ $=17$
$\mathbf{A_2}$	$8(12+3)-[8(5)+14]$ $=66$	$5(12+1)-[5(5)+18]$ $=22$

10.7 Solar Energy Investment Strategy C4–2

	Max	Min
A_1	82	6
A_2	76	34
A_3	97	26
A_4	28	28
A^*_{Maximax}	A_3	
A^*_{Maximin}		A_2

10.8 Restaurant Expansion Strategy C4–2

	Max	Min
A_1	78	12
A_2	68	24
A_3	91	29
A_4	61	37
A^*_{Maximax}	A_3	
A^*_{Maximin}		A_4

10.9 Renewable Energy Investment Strategy C4–2

	Max	Min
A_1	132	-38
A_2	91	6
A_3	65	22
A_4	51	30
A^*_{Maximax}	A_1	
A^*_{Maximin}		A_4